HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and integrated **LLM inference workloads** on IBM Z using **Spyre hardware accelerator cards**, implementing model loading and runtime integration strategies.
- Profiled **LLM inference workloads** to measure latency, throughput, and memory usage, analyzing performance data to identify bottlenecks across hardware utilization and batching strategies.
- Diagnosed and resolved **inference errors** and performance regressions across firmware, drivers, and runtimes, collaborating with hardware engineers and system architects on durable solutions.
- Designed and implemented **monitoring and telemetry** for production LLM workloads, instrumenting systems to capture model performance, hardware utilization, and error rates.
- Participated in architecture reviews and technical discussions across **AI, hardware, firmware, and system software teams**, producing clear technical documentation.
- Wrote **Python scripts** to automate data extraction for model performance analysis, processing approximately **50GB of log data** per week.
- Containerized **LLM inference services** using **Docker** and deployed them to a Kubernetes cluster, managing configuration and resource allocation.

SKILLS

AI/ML – LLM, Transformer Models, Inference Workloads, Performance Profiling, Model Loading, Runtime Integration, Memory Management, Resource Allocation, Quantization, Pruning, Knowledge Distillation

Programming Languages – Python, C/C++, SQL

Systems & Architecture – IBM Z Architecture, Hardware Accelerators (Spyre), Computer Architecture, Memory Hierarchies, Parallel Processing, I/O Systems, Linux Environments, System-Level Debugging

ML Frameworks & Tools – PyTorch, TensorFlow, Hugging Face Transformers, ONNX Runtime, TensorRT, TorchServe

Observability & Operations – Monitoring, Telemetry, Logging, Alerting, Splunk, Dynatrace, Prometheus, Grafana, ELK Stack, Distributed Tracing, OpenTelemetry, Jaeger

Development Practices – API Development, CI/CD Pipelines, Docker, Kubernetes, Git, Code Reviews, Agile Methodologies

PROJECTS

LLM Inference Optimization on Enterprise Hardware

- Engineered a system to load and run transformer-based LLMs on custom hardware accelerators, focusing on efficient memory access patterns and optimized batching strategies.
- Developed Python-based performance profiling tools to capture detailed metrics on latency and throughput, identifying key bottlenecks in the inference pipeline.
- Collaborated with system architects to integrate the LLM runtime with existing enterprise applications through well-defined APIs, enabling new AI capabilities.

Automated Failure Analysis for AI Workloads

- Created Python scripts to parse complex system logs from AI inference runs, identifying root causes of performance regressions and inference errors.
- Implemented automated alerting mechanisms based on log analysis, reducing manual triage time for critical issues by approximately **40%**.
- Contributed to the development of a centralized observability dashboard using **Splunk** to visualize system health and error rates for AI workloads.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)